

Final Report

Analyzing the Relationship Between Daily Habits and Productivity

Motivation

I chose this project because I wanted to see how my own daily habits are connected to my productivity. Instead of just guessing, I thought it would be interesting to actually collect data from my own life and look at the numbers. I was curious about things like — does sleeping more really help? Does spending too much time on my phone hurt my focus? I wanted to find out which habits matter the most and by how much.

Beyond my personal curiosity, I also think this kind of analysis could be useful for other people who have similar routines and struggle with the same things. If the data shows clear patterns, it could help create awareness about everyday choices that we often don't think about.

Data Source

I collected all the data myself, manually, every day for 56 days (March 10 – May 5, 2026). Each evening I filled in a CSV file with my daily measurements. The dataset has 56 rows and 10 columns, including date, sleep hours, sleep quality, morning mood, energy, study hours, phone usage, productivity, stress, and whether it was a weekend.

Most of the variables were measured on a 1–5 scale. I had some basic criteria for each score, but I'll be honest — there was some subjectivity involved, especially for productivity and mood. I tried to be consistent, but since I was rating my own day, personal perception played a role too. This is something to keep in mind when looking at the results.

Data Analysis

Exploratory Data Analysis (EDA)

I started by looking at the basic statistics of the dataset. The average productivity score was around 2.3, which means most days were low to moderate in terms of productivity. I made scatter plots to see how different variables relate to productivity, and a correlation matrix to get a broader picture.

From the correlation analysis, energy had the strongest positive correlation with productivity ($r = 0.29$, $p < 0.05$), and phone usage had a significant negative correlation ($r = -0.29$, $p < 0.05$). I used both Pearson and Spearman tests — Spearman showed slightly stronger results for phone usage and study hours, which makes sense since these variables are ordinal.

I also compared weekday and weekend productivity. The average was a bit higher on weekends (2.69 vs 2.20), though the difference wasn't huge.

Machine Learning Models

I trained four models to predict productivity — two regression models and two classification models. I used an 80/20 train-test split and 5-fold cross-validation to evaluate them.

The Random Forest Regressor performed best among the regression models with $R^2 = 0.59$ and RMSE = 0.68 on the test set. Both classification models (Logistic Regression and Random Forest Classifier) reached 66.7% accuracy, which is reasonable given the five-class nature of the target variable.

The cross-validation scores were lower, especially for regression models (negative R^2), which is mainly because the dataset is very small. With only 56 samples, 5-fold CV leaves very few examples per fold.

Feature importance from the Random Forest model showed that phone usage was the most important predictor, followed by morning mood and sleep hours.

Findings

The main findings from this project are:

Energy and phone usage are the two variables most consistently related to productivity. Energy has a positive effect, while more phone usage is linked to lower productivity — and both are statistically significant.

The machine learning models confirmed this but also added something interesting: phone usage ranked as the top feature in the Random Forest model, even though energy ranked highest in the correlation analysis. This difference suggests that the relationship between these variables and productivity isn't fully linear — tree-based models can pick up on patterns that simple correlations miss.

Weekends tend to be slightly more productive than weekdays, a Mann-Whitney U test confirmed that this difference is not statistically significant ($p = 0.219$), suggesting that weekend status alone does not have a strong impact on productivity.

Overall, the results suggest that if I want to improve my productivity, the two most actionable things are reducing phone usage and making sure my energy levels are good — which probably connects to sleep, exercise, and other lifestyle factors.

Limitations and Future Work

The biggest limitation of this project is the small sample size. 56 days of data is not a lot, and it shows in the model results — especially the cross-validation scores. The models work okay on the test set but don't generalize well, which is expected with this amount of data.

Another issue is that the data is self-reported. My ratings for productivity, mood, and energy are based on my own perception, which can be inconsistent. Some days I might have rated a 3 when it was really a 2, just based on how I felt at that moment.

The class imbalance was also a problem for the classification models. Some productivity levels (like 4 and 5) only appeared a few times, which makes it hard for the model to learn those classes properly.

In the future, it would be interesting to collect data for a longer period maybe 6 to 12 months to

get more reliable results. Adding more variables like exercise, nutrition, or even weather could also improve the analysis. More advanced models like gradient boosting could be tried as well.

AI Usage Disclosure

This project was completed with assistance from Claude. Claude was used to help review and improve code, interpret model outputs, and assist in writing sections of this report. All prompts and outputs were part of an interactive discussion. The analysis, data collection, and final decisions were made by the student.